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| **Fiscal Year:** | FY 2019 | **Task Last Updated:** | FY 07/24/2019 |
| **PI Name:** | Poirier, David Sc.D. | | |
| **Project Title:** | Effect of Varying Convection on Dendrite Morphology and Macrosegregation | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | MATERIALS SCIENCE--Materials science | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **PI Organization Type:** | UNIVERSITY | **Phone:** | 520-621-6072 |
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| **PI Address 1:** | 1235 East James E Rogers Way | | |
| **PI Address 2:** | Department of Materials Science and Engineering | | |
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| **City:** | Tucson | **State:** | AZ |
| **Zip Code:** | 85721-0012 | **Congressional District:** | 7 |
| **Comments:** | | | |
| **Project Type:** | Flight,Ground | **Solicitation / Funding Source:** | 2010 Materials Science NNH10ZTT001N |
| **Start Date:** | 12/01/2014 | **End Date:** | 04/30/2020 |
| **No. of Post Docs:** | 2 | **No. of PhD Degrees:** | 1 |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | 1 |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | 1 |
| **No. of Bachelor's Candidates:** | 1 | **Monitoring Center:** | NASA MSFC |
| **Contact Monitor:** | Grugel, Richard | **Contact Phone:** | 256 544-9165 |
| **Contact Email:** | Richard.N.Grugel@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date change to 4/30/2020 per NSSC information (Ed., 11/13/19) | | |
| **Key Personnel Changes/Previous PI:** | October 2017 report: Prof. D.R. Poirier, Dept. of Materials Science and Engineering, The University of Arizona, remains as the P.I. Dr. V. Manga, Assistant Research Professor, in the Dept. of Materials Science and Engineering at The University of Arizona has been the co-investigator since early 2016. During 2019, Dr. Manga was assisted by a post-doctoral researcher on a temporary basis. Her main responsibility was to introduce Dr. Manga to software (ANSYS-Fluent) for simulating macrosgregation. | | |
| **COI Name (Institution):** | Manga, Venkat Ph.D. ( The University of Arizona )<br>Tewari, Surendra Ph.D. ( Cleveland State University ) | | |
| **Grant/Contract No.:** | NNX15AC04G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | NOTE (Ed., 12/16/2014): This project is a continuation of grant NNX10AV40G, "Effect of Varying Convection on Dendrite Morphology and Macrosegregation," period of performance of 12/1/2010-11/30/2014, and with the same Principal Investigator, Dr. David Poirier.<br>The research of this grant pertains to convection that is driven by directional solidification (DS) through changes in cross-sections (because of solidification shrinkage) and thermosolutal convection. In microgravity, of course, the thermosolutal convection will be mitigated so that effects of the shrinkage-driven flow in cross-sectional changes can be isolated from the usual thermosolutal convection that always occurs when solidification occurs on Earth. The outcome of the research is expected to be relevant to commercial DS-castings, which are used in the high-temperature sections of gas-turbine engines (i.e., jet engines and electric power generators). Such sophisticated castings have geometries, which invariably have changes in cross-sections. Yet convection at a cross-sectional change and its relationship to the formation of defects in DS-castings and/or in dendritic mono-crystals have not been studied well. Hence, the research not only has scientific merit, it is also relevant to the production of commercial DS castings.<br><br>The research is on directional solidification experiments that are compatible with the solidification cartridge assemblies (SCAs) that can be employed in a furnace on the International Space Station (ISS). The research comprises: 1) selected alloys have varying density-responses to gravity forces; 2) crucible designs with variable cross-sections but compatible with the current SCAs; 3) sophisticated characterizations of the solidified microstructures, including both micro- and macro-segregation; and 4) supporting computer simulations. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The outcome of the research is expected to be relevant to commercial directionally solidified-castings, which are used in the high-temperature sections of gas-turbine engines (i.e., jet engines and turbine electric power-generators). Such sophisticated castings have geometries, which invariably have changes in cross-sections. Yet convection at a cross-sectional change and its relationship to the formation of defects in DS-castings and/or in dendritic mono-crystals have not been studied well. Hence, the research is very much needed for its scientific merit and also because it is relevant to the production of commercial DS castings. |
| **Task Progress:** | During this reporting period, we did simulations of the directional solidification of a Pb-Sb alloy through changes in cross-sections. The simulations compare with experimental castings made at Cleveland State University. |
| **Bibliography Type:** | Description: (Last Updated: 06/06/2025) |
| **Articles in Peer-reviewed Journals** | Ghods M, Lauer M, Upadhyay SR, Grugel RN, Tewari SN, Poirier DR. "Spurious grain formation at cross-sectional expansion during directional solidification." Journal of Materials Engineering and Performance. 2018 Jun;27(6):3122-30. https://doi.org/10.1007/s11665-018-3364-0 , Jun-2018 |
| **Articles in Peer-reviewed Journals** | Manga VR, Poirier DR. "Ab initio molecular dynamics simulation of self-diffusion in Al–Si binary melts." Modelling and Simulation in Materials Science and Engineering. 2018 Sep;26(6):065006. https://doi.org/10.1088/1361-651X/aacdbc , Sep-2018 |